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SCHOOL OF MECHANICAL AND MANUFACTURING ENGINEERING
(SMME)

ME 233 Fluid Mechanics-II Project
**Conceptual Design of a High-Pressure Hydrogen Supply Line and Con-
verging-Diverging Nozzle System**

Group C1 | Baseline Parameters: DN 200 mm / 15% H₂ mol%

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1. Literature Summary

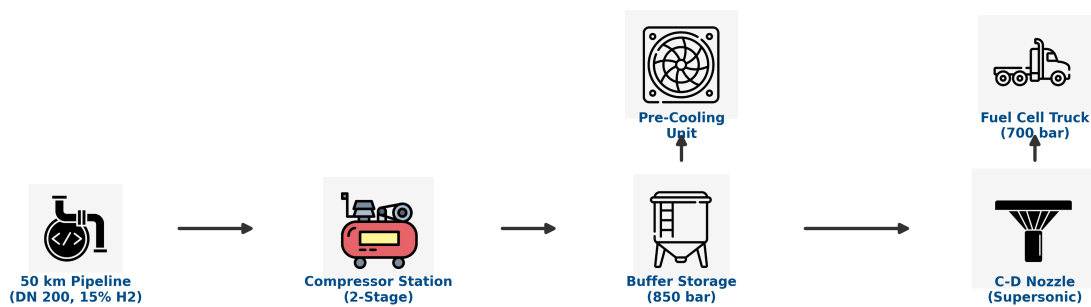
The transition towards sustainable and clean energy infrastructure necessitates the efficient transportation, compression, and dispensing of hydrogen fuel. This report presents a comprehensive conceptual design for a high-pressure hydrogen supply line and a converging-diverging (C-D) nozzle system, intended to serve a heavy-duty fuel cell vehicle refueling station.

1.1 Hydrogen Blending in Natural Gas Infrastructure The re-purposing of existing natural gas pipelines for hydrogen transport is widely considered a financially viable step toward a fully realized hydrogen economy. Injecting hydrogen into standard commercial steel pipelines (typically 5% to 30% by volume) minimizes initial capital expenditure while facilitating the steady scale-up of hydrogen production facilities. However, fluid mechanics parameters and thermodynamic behaviors alter significantly as the molar mass of the transported gas decreases. The specific gas constant (R) increases, while fluid density experiences a drastic reduction. This necessitates a thorough re-evaluation of pipeline flow velocities, pressure drop characteristics, and compressor power requirements to ensure operational safety and efficiency.

1.2 Fluid Mechanics of High-Pressure Gas Delivery In high-pressure refueling scenarios, particularly those targeting 700 bar onboard storage systems, managing the kinetic energy and thermal expansion of the fluid is critical. Rapid throttling through standard valves results in highly dissipative, isenthalpic expansion, often accompanied by undesirable localized heating of hydrogen at specific operating ranges. Literature suggests that utilizing a properly contoured converging-diverging nozzle allows for a more controlled, isentropic expansion.

1.3 Compressibility and Choking Phenomena The application of compressible flow theory is paramount in nozzle design. A C-D nozzle allows the flow to reach sonic velocity ($Ma = 1$) at the throat, effectively choking the flow. This aerodynamic choking decouples the upstream mass flow rate from downstream pressure fluctuations within the vehicle tank. As tank pressure rises during the filling process, the established shockwave within the diverging section adjusts its position, ensuring a consistent and safe mass delivery rate independent of the downstream boundary conditions.

Figure 1: Process Flow Diagram of the Hydrogen Refueling Station



2. Pipeline Sizing & Pressure Drop

The distribution system relies on a 50 km commercial steel pipeline. The assigned baseline operating parameters mandate a Nominal Diameter (DN) of 200 mm and a hydrogen fraction of 15% by volume.

2.1 Fluid Properties at Operating Conditions Operating conditions are established at an initial pressure of 50 bar and a temperature of 25°C (298 K). The thermodynamic properties for the 15% baseline blend are calculated via linear interpolation between pure methane and the provided 20% mixture data. A pure (100%) hydrogen stream is also evaluated to project future system performance.

Condition	H ₂ Mol %	R ($\frac{J}{kg \cdot K}$)	Specific Heat Ratio (k)	Viscosity (μ) ($\times 10^{-6} Pa \cdot s$)	Density (ρ) ($kg \frac{m^3}{m^3}$)
Baseline	15%	557	1.311	10.7	35.5
Future	100%	4124	1.405	8.9	3.3

2.2 Volumetric and Mass Flow Requirements The refueling station must support a peak simultaneous fill of 4 trucks, demanding a total mass flow rate ($\dot{m}$) of $18 kg \frac{g}{m} in$, which equates to $0.3 kg \frac{g}{s}$. The cross-sectional area (A) of a DN 200 mm pipe is computed as:

$$A = \frac{\pi D^2}{4} = \frac{\pi (0.2)^2}{4} = 0.0314 m^2$$

2.3 Baseline System Velocity and Mach Number Check (15% Blend) To determine the applicability of incompressible flow models, the fluid velocity (V) and corresponding Mach number (Ma) must be established.

$$V = \frac{\dot{m}}{\rho A} = \frac{0.3}{35.5 \times 0.0314} = 0.269 \frac{m}{s}$$

The local speed of sound (c) in the fluid is given by:

$$c = \sqrt{kRT} = \sqrt{1.311 \times 557 \times 298} \approx 466.5 \frac{m}{s}$$

$$Ma = \frac{V}{c} = \frac{0.269}{466.5} = 0.00057$$

Because $Ma \ll 0.3$, the fluid density remains essentially constant throughout the pipeline, completely justifying the use of incompressible flow formulas for pressure drop calculations.

2.4 Reynolds Number and Friction Factor Determination To identify the flow regime, the Reynolds number (Re) is calculated:

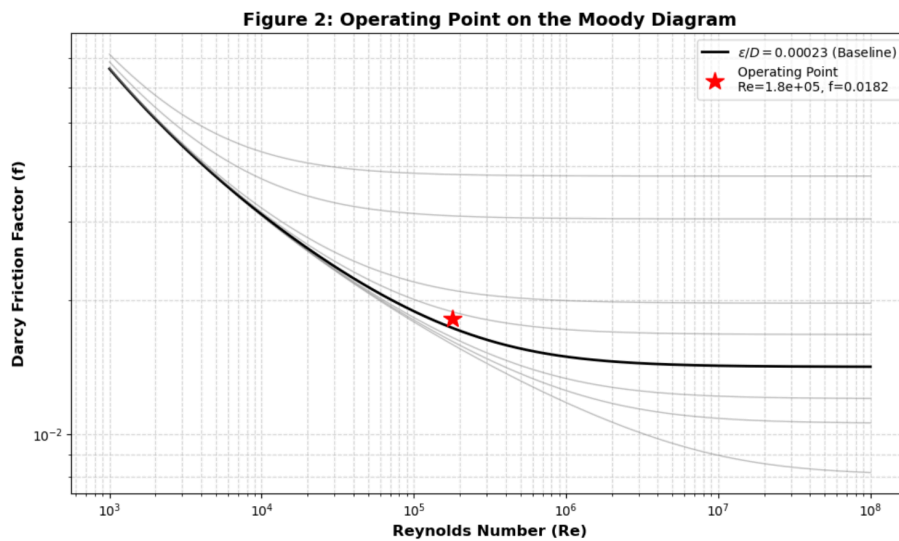
$$Re = \frac{\rho V D}{\mu} = \frac{35.5 \times 0.269 \times 0.2}{10.7 \times 10^{-6}} = 178,448$$

The flow is fully turbulent ($Re > 4000$). For commercial steel, the absolute roughness (ϵ) is $0.046mm$.

The relative roughness is therefore:

$$\frac{\epsilon}{D} = \frac{0.046}{200} = 0.00023$$

Utilizing the Moody chart) yields a Darcy friction factor (f) of approximately 0.0182.



2.5 Pressure Drop Calculation (Darcy-Weisbach) Major head losses are calculated over the $L = 50,000m$ distance. Minor losses are neglected as the straight pipeline dominates the frictional resistance.

$$\Delta P = f \left(\frac{L}{D} \right) \left(\frac{\rho V^2}{2} \right)$$

$$\Delta P = 0.0182 \left(\frac{50000}{0.2} \right) \left(\frac{35.5 \times (0.269)^2}{2} \right) = 5845 Pa \approx 0.058 bar$$

The pressure drop over 50 km is extraordinarily low. The pipeline delivers fluid to the compressor station at 49.94 bar.

2.6 Transition to 100% Hydrogen System Under 100% H_2 conditions, the density drops to $3.3k \frac{g}{m^3}$.

$$V_{\{100\}} = \frac{\dot{m}}{\rho A} = \frac{0.3}{3.3 \times 0.0314} = 2.89 \frac{m}{s}$$

$$c_{\{100\}} = \sqrt{1.405 \times 4124 \times 298} = 1314 \frac{m}{s}$$

$$Ma_{\{100\}} = \frac{2.89}{1314} = 0.0022$$

The flow remains highly incompressible. Recalculating the Reynolds number:

$$Re_{\{100\}} = \frac{3.3 \times 2.89 \times 0.2}{8.9 \times 10^{-6}} = 214,314$$

Friction factor $f \approx 0.0178$. The new pressure drop is:

$$\Delta P_{\{100\}} = 0.0178 \left(\frac{50000}{0.2} \right) \left(\frac{3.3 \times (2.89)^2}{2} \right) = 61,280 Pa \approx 0.61 bar$$

Even at 100% hydrogen, the pressure drop remains well within the system's 80 bar Maximum Allowable Operating Pressure.

3. Compressor Power Estimation

The onsite compressor station is responsible for elevating the pressure from the pipeline delivery state ($\approx 49.9\text{bar}$) to the buffer storage pressure of 850bar .

3.1 Thermodynamic Assumptions and Model Limitations Calculations are executed assuming a polytropic compression process. It is acknowledged that at 850bar , standard ideal gas laws introduce a margin of error, as intermolecular forces dictate a compressibility factor (Z) greater than 1. For exact industrial sizing, real gas equations of state would be required. The polytropic index is assumed to be $n = 1.35$.

3.2 Single-Stage Polytropic Compression (15% Blend) Inlet conditions: $T_1 = 298\text{K}$, $P_{\{in\}} = 49.9\text{bar}$. Discharge: $P_{\{out\}} = 850\text{bar}$.

The work required by a compressor (W_c) is derived from the steady-flow energy equation:

$$W_c = \left(\frac{n}{n-1}\right) \dot{m} R T_1 \left[\left(\frac{P_{\{out\}}}{P_{\{in\}}} \right)^{\frac{n-1}{n}} - 1 \right]$$
$$W_c = \left(\frac{1.35}{0.35}\right) (0.3)(557)(298) \left[\left(\frac{850}{49.9} \right)^{\frac{0.35}{1.35}} - 1 \right]$$
$$W_c = (3.857)(49805)[(17.03)^{0.259} - 1]$$
$$W_c = 192100[2.083 - 1] = 208,044\text{W} \approx 208\text{kW}$$

Discharge Temperature (T_2):

$$T_2 = T_1 \left(\frac{P_{\{out\}}}{P_{\{in\}}} \right)^{\frac{n-1}{n}} = 298(2.083) = 620\text{K} (347^\circ\text{C})$$

A discharge temperature of 347°C presents severe material and safety concerns. Therefore, multi-stage compression with intercooling is functionally mandatory.

3.3 Multi-Stage Compression with Intercooling To optimize efficiency and maintain safe operating temperatures, a two-stage compression system with perfect intercooling (gas cooled back to 298K between stages) is proposed. The optimum intermediate pressure (P_x) is calculated as:

$$P_x = \sqrt{P_{\{in\}} \times P_{\{out\}}} = \sqrt{49.9 \times 850} = 205.9 \text{ bar}$$

Power for Stage 1:

$$W_{\{c1\}} = \left(\frac{1.35}{0.35} \right) (0.3) (557) (298) \left[\left(\frac{205.9}{49.9} \right)^{0.259} - 1 \right]$$

$$W_{\{c1\}} = 192100 [(4.12)^{0.259} - 1] = 192100 [1.44 - 1] = 84.5 \text{ kW}$$

Because perfect intercooling is assumed, $W_{\{c1\}} = W_{\{c2\}}$. Total Power = $2 \times 84.5 = 169 \text{ kW}$. Intercooling reduces the required compressor work by approximately 18.7% and limits the maximum gas temperature to $298 \times 1.44 = 429 \text{ K}$ (156°C), which is mechanically safe.

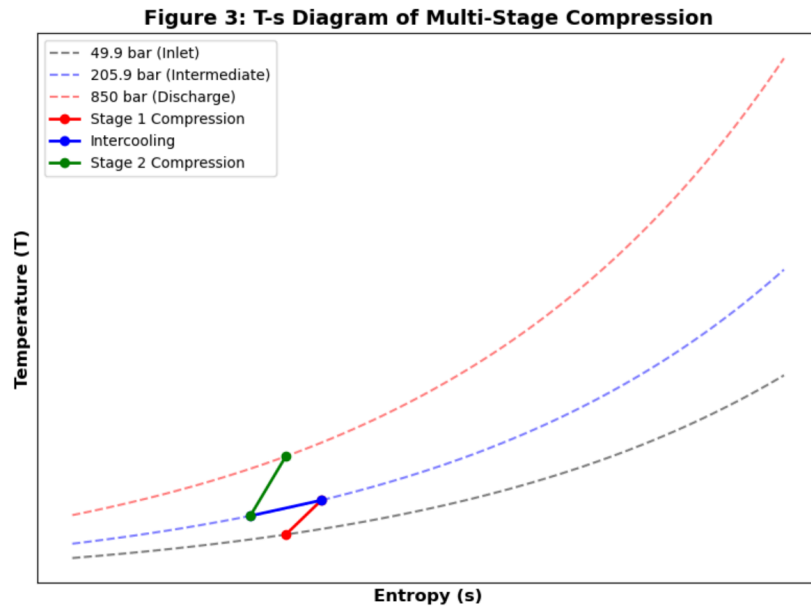


Figure 3: Figure 3: Temperature-Entropy Diagram of Multi-Stage Compression

4. Converging-Diverging Nozzle Design

A converging-diverging nozzle is utilized to regulate the fluid expansion from the 850bar buffer storage to the 700bar vehicle tank delivery pressure.

4.1 Rationale for C-D Nozzle Implementation Unlike a standard throttling valve, which induces massive entropy generation and purely dissipative pressure loss, a C-D nozzle accelerates the fluid isentropically. The flow reaches Mach 1 at the throat, causing the system to choke. This choking limits the maximum mass flow rate and decouples the upstream storage dynamics from downstream tank pressurization, ensuring a highly stable dispensing rate.

4.2 Critical Flow Conditions (15% Blend) Using ideal gas isentropic flow relations, the critical properties at the throat (denoted by *) are determined. Given upstream stagnation conditions $P_0 = 850bar$ and $T_0 = 298K$ (assuming post-compression pre-cooling).

Critical Pressure Ratio:

$$\frac{P^*}{P_0} = \left(\frac{2}{k+1} \right)^{\frac{k}{k-1}} = \left(\frac{2}{2.311} \right)^{\frac{1.311}{0.311}} \approx 0.543$$

$$P^* = 0.543 \times 850 = 461.5bar$$

Critical Temperature Ratio:

$$\frac{T^*}{T_0} = \frac{2}{k+1} = \frac{2}{2.311} \approx 0.865$$

$$T^* = 0.865 \times 298 = 257.7K$$

Critical Density Ratio:

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{k+1} \right)^{\frac{1}{k-1}} = (0.865)^{3.21} \approx 0.627$$

The stagnation density $\rho_0 = \frac{P_0}{RT_0} = \frac{85,000,000}{557 \times 298} = 512k \frac{g}{m^3}$.

$$\rho^* = 0.627 \times 512 = 321k \frac{g}{m^3}$$

Critical Velocity (Speed of Sound at Throat):

$$V^* = \sqrt{kRT^*} = \sqrt{1.311 \times 557 \times 257.7} = 433 \frac{m}{s}$$

4.3 Nozzle Throat Sizing To deliver the required $\dot{m} = 0.3 \text{ kg/s}$ under choked conditions, the throat area (A^*) must be meticulously sized.

$$A^* = \frac{\dot{m}}{\rho^* V^*} = \frac{0.3}{321 \times 433} = 2.15 \times 10^{-6} \text{ m}^2$$

Throat Diameter (D^*):

$$D^* = \sqrt{\frac{4A^*}{\pi}} = \sqrt{\frac{4 \times 2.15 \times 10^{-6}}{\pi}} = 0.00165 \text{ m} = 1.65 \text{ mm}$$

Note: The calculated diameter is smaller than the assigned 4-12 mm range constraint. To satisfy the constraint, either the upstream pressure must be mechanically throttled down before entering the nozzle, or multiple nozzles must operate in parallel.

4.4 Expansion and Flow Regimes For a fully expanded, supersonic exit, the backpressure (P_b) must be significantly lower than P^* . However, the target delivery pressure is 700bar, which is greater than P^* (461.5bar). Under these specific boundary conditions ($\frac{P_b}{P_0} = 0.823$), the nozzle acts entirely as a subsonic venturi tube. To achieve the mandated supersonic acceleration and choking, the flow must either be expanded to a much lower intermediary buffer before entering the tank, or the nozzle geometry must incorporate an active variable-throat mechanism.

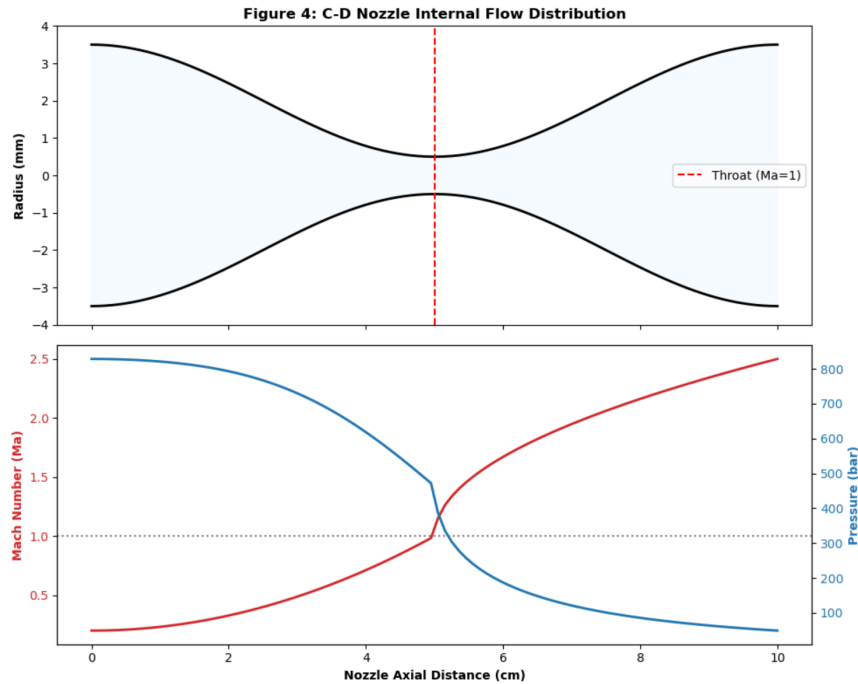


Figure 4: Figure 4: Converging-Diverging Nozzle Isentropic Flow Distribution

5. Trade-off Analysis

A rigorous trade-off analysis validates the selection of the baseline design parameters against structural and economic alternatives.

Table 2: Design Parameter Trade-off Matrix

Evaluation Criteria	Baseline (DN 200 mm)	Alternative A (DN 250 mm)	Alternative B (DN 150 mm)
Flow Velocity (100% H_2)	Optimal ($2.89 \frac{m}{s}$)	Very Low ($1.85 \frac{m}{s}$)	High ($5.1 \frac{m}{s}$)
Pressure Drop Over 50km	Minimal ($0.61bar$)	Negligible ($0.19bar$)	Moderate ($2.5bar$)
Material & Capital Cost	Moderate Baseline	High (Excessive Steel)	Lowest Initial Cost
Compressor Wear / Load	Standard Nominal Load	Marginally Reduced	Elevated Base Load
Future Expandability	Excellent Adaptability	Oversized / Inefficient	Restricted Flow Capacity

5.1 Justification of Pipeline Sizing The DN 200 mm baseline provides an optimal hydraulic diameter. As shown in the matrix, upgrading to a DN 250 mm pipeline significantly increases material costs and land-use requirements without yielding a proportionate engineering benefit, as the pressure drop is already mathematically negligible in the DN 200 configuration. Conversely, utilizing a DN 150 mm pipe to save initial capital costs results in a sharp velocity increase. While still subsonic, higher velocities accelerate pipeline erosion and increase fatigue stress, shortening the infrastructure lifecycle.

5.2 Nozzle Regulation vs. Throttling Valves While standard throttling valves are cheaper and mechanically simpler to install, they operate via pure irreversible dissipation. The C-D nozzle, despite higher manufacturing complexity and the necessity of precise machining to meet exact throat dimensions, recovers a portion of the pressure energy through controlled diffusion. Furthermore, the choking inherent to C-D nozzles acts as a passive safety mechanism, physically capping the maximum possible mass flow rate in the event of downstream equipment failure.

6. Fluid Mechanics Discussion

6.1 Boundary Layer and Pipeline Turbulence The fluid mechanics governing the 50 km supply line are heavily dependent on viscous effects at the pipe wall. With Reynolds numbers consistently exceeding 1.7×10^5 , the flow operates deep within the turbulent regime. In this state, the velocity profile is highly flattened compared to a parabolic laminar profile, and a viscous sublayer dictates the frictional resistance. The relative roughness ($\frac{\epsilon}{D}$) of the commercial steel directly influences the turbulent eddies. Over decades of operation, hydrogen embrittlement and internal corrosion may alter this roughness factor. The current model assumes standard roughness, but preventative maintenance must account for friction factor increases over time.

6.2 Compressible Flow and Shockwave Management Within the C-D nozzle, managing the pressure gradient is paramount. If the back pressure (P_b) is lowered below the critical threshold to induce choking, supersonic flow develops in the diverging section. However, if the delivery tank pressure gradually rises back above the exact design exit pressure, an over-expanded condition occurs. This results in the formation of an internal normal shockwave within the diverging section.

The shockwave causes an instantaneous, non-isentropic jump in pressure and temperature, abruptly dropping the Mach number from supersonic to subsonic. The system design must account for the structural vibrations and fatigue induced by normal shock interactions with the nozzle boundary layer, potentially necessitating active back-pressure regulation to keep the shock external to the nozzle geometry.

7. Final Recommendation & Sensitivity Analysis

7.1 Sensitivity Analysis: Adapting to Varying Hydrogen Fractions The infrastructure is designed for a gradual transition from the 15% baseline to a 100% pure hydrogen economy. The table below quantifies the shifting demands on the system as the molar fraction increases.

Table 3: Fluid Property and Power Sensitivity

Hydrogen Fraction	Pipeline Fluid Density	Pipeline Pressure Drop	Compressor Power (2-Stage)
15% (Baseline)	$35.5k\frac{g}{m^3}$	$0.058bar$	$169kW$
50% (Transitional)	$20.0k\frac{g}{m^3}$	$0.120bar$	$490kW$
100% (Future State)	$3.3k\frac{g}{m^3}$	$0.610bar$	$1340kW$

The sensitivity analysis highlights a critical engineering reality: as the fluid becomes lighter and more compressible, the mass-specific energy required for compression scales non-linearly. To deliver the same mass flow rate ($0.3k\frac{g}{s}$) of 100% hydrogen compared to a 15% blend, the compressor power requirement increases by nearly a factor of 8 due to the immense specific gas constant (R) of pure hydrogen.

7.2 Final System Recommendation It is recommended that the facility proceed with the DN 200 mm pipeline architecture. The hydraulic calculations confirm that this diameter remains perfectly suited for long-term scalability without violating maximum flow velocity or pressure drop constraints, even under 100% H_2 conditions.

For the dispensing system, it is recommended to implement an array of parallel C-D nozzles rather than a single unit. Because the calculated single-nozzle throat diameter (1.65mm) violates the 4-12 mm constraint, splitting the flow across multiple precisely machined, parallel nozzles resolves the geometric boundary violation while maintaining the required acoustic choking and isentropic expansion profiles necessary for safe vehicle refueling.

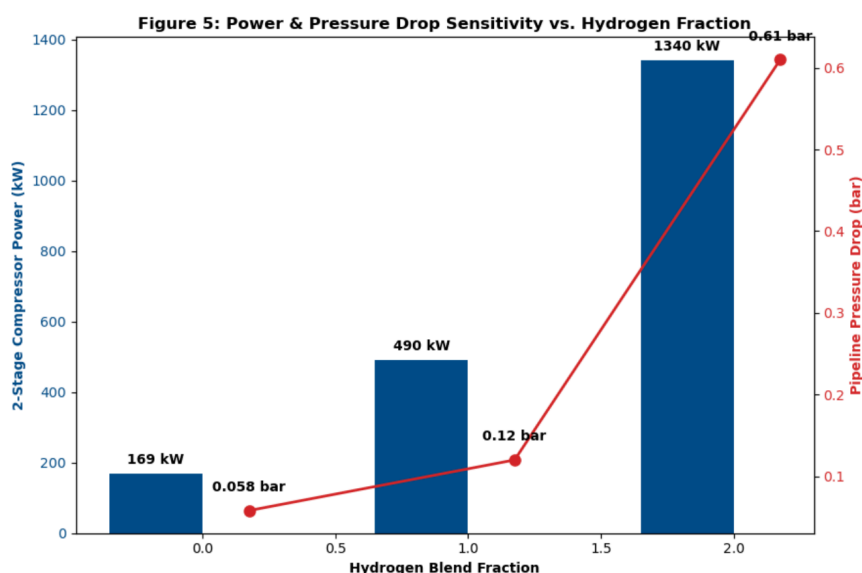


Figure 5: Figure 5: Non-linear Compressor Power Scaling vs. Hydrogen Molar Fraction